

# Taichi Kamei

t.kamei0716@gmail.com | Portfolio | [github.com/Taichi-Kamei](https://github.com/Taichi-Kamei) | [linkedin.com/in/taichikamei](https://linkedin.com/in/taichikamei)

## EDUCATION

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### 3rd Year Engineering Physics

University of British Columbia | Vancouver

*Expected Graduation - May 2028*

Relevant Courses: Solid and Fluid Mechanics, Mechanical Designs, Electric Circuit Analysis, Signals and Systems, Thermodynamics, Statistical Mechanics, Quantum Mechanics

Dean's Honor List 2025

## EXPERIENCE

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### Firmware & Validation Engineer Co-op

UBC Blusson Quantum Matter Institute

*May 2026 - Present*

- Developing full-stack C++ firmware for 4-channel low-noise SNSPD current source, achieving lower noise spectral density than commercial SMU
- Implementing 24-bit ADC and 16-bit DAC SPI driver on ESP32 using ESP-IDF, enabling precision current measurement and control
- Designing FSM with FreeRTOS dual-core task architecture and ISR-driven sampling
- Developing automated V-I sweep to characterize SNSPD switching current and determine optimal bias point
- Building Python Qt6 GUI and CSV data logging for instrument and data acquisition
- Testing and verifying the noise floor, negative rail voltage noise, time domain voltage drift characteristics, and DAC&ADC accuracy of the instrument using oscilloscope and pico-ammeter
- Designing a current source PCB casing on SolidWorks, reducing vibrational noise coupling into the PCB

## PROJECTS

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### Drone Flight Controller

*Feb 2026 - Present*

- Designed a power board for 4S LiPo, integrating BMS IC and 5V/5A buck converter
- Designed a custom 4-layer GPS & magnetometer PCB on KiCAD fitting within 260mm by 260mm
- Prototyping a flight controller PCB, integrating ESP32-S3-Mini-1U, BMS, IMU, barometer, and a RF transceiver using SX1261

### Autonomous Clue Detecting Robot

*November 2025*

- Developed an autonomous vehicle in Gazebo and ROS 1 using OpenCV for PID control driving, obstruction avoidance, and CNN clue detection
- Designed and implemented a finite state machine (FSM) capable of transitioning its states for moving obstruction detection, clue board detection, and drive under various track surface conditions
- Optimized a PID algorithm for precise center-lane driving using dual side lines instead of a typical single line-following, overcoming camera center offset from the geometric center of the lane
- Created a Qt5 controller GUI which integrates simulation controls and launch of two ROS node scripts, boosting team productivity by centralizing all controls into a single window which normally takes 4+ separate terminals

### Autonomous Payload Retrieving Robot

*June - August 2025*

- Designed a 3-DOF four-bar-linkage arm and arm base on Onshape and fabricated components using 3D printer, water jet cutter, and laser cutter
- Sized appropriate servo motors for the arm from maximum payload and bending moment calculations
- Implemented magnetic encoder-based rotational control for the arm base, achieving accuracy of less than 1° error
- Designed a custom 2-layer I2C multiplexer and I2C buffer PCB on KiCAD to solve peripheral address conflicts and signal degradation
- Prototyped a payload detection algorithm in Python on Raspberry Pi using two 2D LiDAR, generating depth and reflectance map in 7Hz. Implemented it in C++ on ESP32 with 15 Hz real-time detection

## SKILLS

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|-------------------|------------------------------------------------------------------------------------------|
| <b>Software</b>   | Python, Java, C/C++, Assembly, VHDL, Linux, Git                                          |
| <b>Embedded</b>   | ESP-IDF, FreeRTOS, Arduino, Raspberry Pi, FPGA, ROS, I2C, SPI, UART                      |
| <b>Electrical</b> | Kicad, Altium, LTSpice, Soldering, Oscilloscope, DMM, Spectrum Analyzer, Logic Analyzer  |
| <b>Mechanical</b> | Onshape, Siemens NX, 3D printing, Laser Cutting, Water Jet Cutting, Drill Press, Caliper |